Spectrum and propagation

$$\lambda = \frac{c}{f\sqrt{\varepsilon_{eff}}}, \qquad c = 3 \times 10^8 \text{ m/s}$$

Band	Range	Aviation use
LF / MF	30-300 kHz / 0.3-3 MHz	NDB
HF	3-30 MHz	oceanic voice (sky wave)
VHF	30-300 MHz	VOR/ILS 108-118, voice 118-137 MHz
UHF	0.3-3 GHz	DME, SSR, GPS (L band)
SHF	3-30 GHz	radars

Ground wave 20 kHz-50 MHz (follows the Earth, less loss over sea, vertical pol.); space wave > 50 MHz (line of sight + ground reflection); sky wave HF (ionosphere). Ionosphere: D absorbs (day only); E, F1, F2 reflect up to about 5, 10, 30 MHz.

$$\text{NDB range [nmi]} \approx 3\sqrt{P\,[\text{W}]}\ \text{(sea)}, \qquad 2\sqrt{P\,[\text{W}]}\ \text{(land)}$$

$$d\,[\text{nmi}] = 1.23\left(\sqrt{H_T\,[\text{ft}]} + \sqrt{H_R\,[\text{ft}]}\right), \qquad 1\text{ ft} = 0.3048\text{ m}$$

4x higher → 2x farther

$$f_c\,[\text{Hz}] = 9\sqrt{N\,[\text{m}^{-3}]}, \qquad f_c\,[\text{MHz}] = 9 \times 10^{-3}\sqrt{N\,[\text{cm}^{-3}]}$$

$$\text{MUF} = \frac{f_c}{\sin\theta}, \quad \text{OWF} = 0.85\,\text{MUF},$$

$$\frac{\text{MUF}_{\text{day}}}{\text{MUF}_{\text{night}}} = \sqrt{\frac{N_{\text{day}}}{N_{\text{night}}}} \approx 2$$

LUF: lowest usable, set by D-layer absorption

Reflection: obstacle >> λ. Refraction: change of medium. Diffraction: edges (behind hills). Scattering: obstacles << λ (rain). Absorption peaks: H2O 22 and 183 GHz, O2 60 and 119 GHz (60 GHz: secure short links); windows 35 and 94 GHz.

Modulation and receivers

$$x_{AM}(t) = A_C\,[1 + m\,x(t)]\cos(2\pi f_C t), \qquad m = \frac{A_m}{A_C},\ 0 < m < 1$$

$$A_C\cos\omega_C t + \tfrac{mA_C}{2}\cos(\omega_C - \omega_m)t + \tfrac{mA_C}{2}\cos(\omega_C + \omega_m)t,$$

$$B_{DSB} = 2f_m$$

$$x_{FM}(t) = A_C\cos\left[2\pi f_C t + k_f\int x\,d\tau\right], \qquad m_f = \frac{\Delta f}{f_m}$$

$$B_{FM} = 2(\Delta f + f_m) = 2f_m(m_f + 1),$$

$$x_{FM} = A_C\sum_n J_n(m_f)\cos 2\pi(f_C + nf_m)t$$

ASK: '1' carrier, '0' nothing. FSK: f1 / f2. PSK: '0' shifts the phase 180°. FDM: frequency slots (channel + guard band); TDM: time slots on the whole band. I/Q: bit pairs on cos and sin of one carrier.

$$f_{IF} = |f_{RF} - f_{LO}|,$$

$$f_{im} = f_{LO} \pm f_{IF} \quad (\text{high-side LO: } f_{im} = f_{RF} + 2f_{IF})$$

high-side LO: f_im = f_RF + 2 f_IF; low-side: f_im = f_RF - 2 f_IF; the RF filter must reject it

Decibels

$$G\,[\text{dB}] = 10\log\frac{P_2}{P_1}, \quad P\,[\text{dBm}] = P\,[\text{dBW}] + 30,$$

$$V\,[\text{dB}\mu\text{V}] = V\,[\text{dBV}] + 120$$

$$20\log\frac{V_2}{V_1}\ (\text{equal } Z), \qquad 20\log\frac{V_2\sqrt{Z_1}}{V_1\sqrt{Z_2}}$$

dB + dB = dB; dBm + dB = dBm; NEVER dBm + dBm: 30 dBm + 30 dBm = 2 W = 33 dBm. Cascade gain: G_T [dB] = sum G_i - sum L_j.

Noise

$$V_{n,\text{rms}} = \sqrt{4kTBR}, \qquad N = kTB, \qquad k = 1.38 \times 10^{-23}\text{ J/K}$$

$$kT_0 = 4.0 \times 10^{-21}\text{ W/Hz} = -204\text{ dBW/Hz} = -174\text{ dBm/Hz}$$

$$(T_0 = 290\text{ K})$$

$$F = \frac{\text{SNR}_{in}}{\text{SNR}_{out}} \geq 1,$$

$$NF = 10\log F, \quad T_e = T_0(F - 1), \quad N_o = kBG(T_0 + T_e)$$

$$F_T = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 G_2} + \cdots$$

passive lossy part at T0 with loss L: G = 1/L, F = L (NF [dB] = L [dB])

$$N_o\,[\text{dBm}] = -174 + NF + 10\log B + G$$

$$P_{\min} = kT_0 BF\,\text{SNR}_{\min}$$

$$\Rightarrow \quad P_{\min}\,[\text{dBm}] = -174 + NF + 10\log B + \text{SNR}_{\min}$$

improve it with a lower NF (LNA first), a narrower B, a more robust modulation

Linearity and dynamic range

$$v_o = \alpha_0 + \alpha_1 v_i + \alpha_2 v_i^2 + \alpha_3 v_i^3 + \cdots, \qquad G_{\text{fund}} = \alpha_1 + \tfrac{3}{4}\alpha_3 A^2$$

One tone: harmonics 2f0, 3f0 (filterable). Two tones: IM3 at 2f1 - f2 and 2f2 - f1, IN band. IM3 grows 3 dB per dB of input.

$$\text{OIP3} = \text{IIP3} + G, \qquad P_{1\text{dB}} \approx \text{IIP3} - 9.6\text{ dB}$$

$$P_{IM3,in} = 3P_{in} - 2\,\text{IIP3}, \qquad \frac{C}{IM3} = 2(\text{IIP3} - P_{in})$$

$$\frac{1}{\text{IIP3}_T} = \frac{1}{\text{IIP3}_1} + \frac{G_1}{\text{IIP3}_2} + \frac{G_1 G_2}{\text{IIP3}_3} + \cdots$$

convert OIP3 to IIP3 first; the dominant term is often a LATE stage

$$DR_o = 1\text{dBCP}_o - N_o, \quad DR_i = 1\text{dBCP}_i - P_{\min},$$

$$1\text{dBCP}_i = 1\text{dBCP}_o - (G - 1)$$

Transmission lines

$$\lambda_g = \frac{c}{f\sqrt{\varepsilon_{eff}}}, \qquad \text{line if } l \gtrsim \lambda_g/10$$

$$V(x) = V^+e^{-\gamma x} + V^-e^{\gamma x}, \quad \gamma = \sqrt{(R + j\omega L)(G + j\omega C)} = \alpha + j\beta,$$

$$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}}$$

$$\text{lossless: } Z_0 = \sqrt{\frac{L}{C}}, \quad \beta = \omega\sqrt{LC} = \frac{2\pi}{\lambda_g},$$

$$v_p = \frac{1}{\sqrt{LC}}, \quad Z_0 = \frac{1}{v_p C} = Lv_p$$

$$\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0}, \qquad \Gamma(-l) = \Gamma_L e^{-j2\beta l}$$

$$Z_{in} = Z_0\frac{Z_L + jZ_0\tan\beta l}{Z_0 + jZ_L\tan\beta l}$$

$$Z_{in}^{sc} = jZ_0\tan\beta l, \quad Z_{in}^{oc} = -jZ_0\cot\beta l, \quad Z_{in}^{\lambda/4} = \frac{Z_0^2}{Z_L}, \quad Z_{in}^{\lambda/2} = Z_L$$

$$\text{SWR} = \frac{1 + |\Gamma|}{1 - |\Gamma|}, \qquad |\Gamma| = \frac{\text{SWR} - 1}{\text{SWR} + 1}$$

$$P_r = |\Gamma_L|^2 P_{in}, \quad P_t = (1 - |\Gamma_L|^2) P_{in},$$

$$RL = -20 \log |\Gamma_L|, \quad IL = -10 \log \frac{P_t}{P_{in}}$$

SWR 1.5 → |Γ| = 0.2, 4 % reflected, RL 14 dB; 10 % reflected = SWR 1.92 = RL 10 dB

Coaxial cable

$$Z_0 = \frac{138}{\sqrt{\varepsilon_r}} \log_{10} \frac{D}{d}, \qquad VF = \frac{v_p}{c} = \frac{1}{\sqrt{\varepsilon_r}}$$

PE ε_r 2.3 → VF 66 %; PTFE 2.1 → 69 %

$$C = \frac{24.12\,\varepsilon_r}{\log_{10}(D/d)} = \frac{3326\sqrt{\varepsilon_r}}{Z_0}\ \mathrm{pF/m}, \qquad Z_0 = \frac{1}{v_p C}$$

$$A = \frac{1.43R}{Z_0} + 9.15\sqrt{\varepsilon_r}\, f\, F\ \ [\mathrm{dB}/100\ \mathrm{m}], \qquad R = 25.4 \left(\frac{1}{d} + \frac{1}{D}\right) \sqrt{f}$$

d, D in mm; f in MHz; F = tan(δ)

Smith chart and S-parameters

$$z = \frac{Z}{Z_0} = \frac{1+\Gamma}{1-\Gamma}, \qquad \Gamma = \frac{z-1}{z+1}, \qquad y = \frac{1}{z}\ \text{(opposite point)}$$

Centre: matched. Left: short (Γ = -1). Right: open (Γ = +1). Upper half inductive, lower capacitive. Towards the generator: CLOCKWISE on the |Γ| circle, one turn = $\lambda_g/2$ (2 β l = 720° per λ). SWR = r where the circle cuts the right real axis. Normalise before, de-normalise after.

$$\begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}, \qquad S_{ij} = \left.\frac{b_i}{a_j}\right|_{a_k=0}$$

a = V+/sqrt(Z0), b = V-/sqrt(Z0); RL = -20 log|S11|; gain or IL from |S21|^2

Matching

$$R_L > Z_0:\ B = \frac{X_L \pm \sqrt{R_L/Z_0}\sqrt{R_L^2 + X_L^2 - Z_0 R_L}}{R_L^2 + X_L^2},$$

$$X = \frac{1}{B} + \frac{X_L Z_0}{R_L} - \frac{Z_0}{B R_L}$$

$$R_L < Z_0:\ X = \pm\sqrt{R_L(Z_0 - R_L)} - X_L, \qquad B = \pm\frac{1}{Z_0}\sqrt{\frac{Z_0 - R_L}{R_L}}$$

$$X > 0: L = \frac{X}{\omega}, \quad X < 0: C = \frac{1}{\omega |X|},$$

$$B > 0: C = \frac{B}{\omega}, \quad B < 0: L = \frac{1}{\omega |B|}$$

$$Z_1 = \sqrt{Z_0 Z_L}, \quad l = \frac{\lambda_g}{4},$$

$$\frac{\Delta f}{f_0} = 2 - \frac{4}{\pi} \arccos\left[\frac{\Gamma_m}{\sqrt{1-\Gamma_m^2}} \frac{2\sqrt{Z_0 Z_L}}{|Z_L - Z_0|}\right]$$